

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
IILM UNIVERSITY, GREATER NOIDA

A Project Report On

HYBRID INTRUSION DETECTION SYSTEM USING ANOMALY AND SIGNATURE-BASED TECHNIQUES

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May, 2026

Candidate's Declaration

We hereby declare that the project entitled "Hybrid Intrusion Detection System Using Anomaly and Signature-Based Techniques" is our original work, carried out from August 2025 to May 2026, under the supervision of Dr. Narinder Verma, Assistant Professor, IILM University, Greater Noida. This project has not been submitted for any other degree or examination.

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Certificate

This is to certify that the project entitled "Hybrid Intrusion Detection System Using Anomaly and Signature-Based Techniques" submitted by Akshat Shakya, Kushal Sharma, Ruprekha Behera, Shivam, and Vivek Chaudhary is a record of original work carried out under my supervision in partial fulfilment of the requirement for the degree of Bachelor of Technology in Computer Science and Engineering, IILM University, Greater Noida. This work has not been submitted elsewhere for any degree.

Signature of HoD: _____ Dr. Anil Saroliya, Professor IILM University, Greater Noida	Signature of Guide: _____ Dr. Narinder Verma, Assistant Professor IILM University, Greater Noida
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Abstract

Traditional Intrusion Detection Systems either miss new (zero-day) attacks or produce too many false alarms, making them unreliable in real-world use.

This project proposes a Hybrid Intrusion Detection System (HIDS) that combines rule-based signature matching with a machine learning model (Random Forest and XGBoost) to detect both known and unknown network threats. A confidence threshold is applied to reduce false alarms.

The system was tested on the UNSW-NB15 dataset and achieved 94.6% accuracy with a false positive rate of just 1.2%, significantly outperforming standalone approaches.

Keywords: Intrusion Detection System, Hybrid IDS, Machine Learning, Random Forest, XGBoost, Network Security, False Positive Reduction.

Table of Contents

Section	Page
Declaration	ii
Certificate	iii
Acknowledgement	iv
Abstract	v
Table of Contents	vi
CHAPTER 1: INTRODUCTION	
1.1 Introduction	
1.2 Problem Statement	
1.3 Objectives	
1.4 Scope of Research	
CHAPTER 2: LITERATURE REVIEW	
2.1 Background	
2.2 Review of Related Work	
2.3 Research Gaps	
CHAPTER 3: METHODOLOGY	
3.1 System Overview	
3.2 Data Used	
3.3 Data Preprocessing	
3.4 Signature-Based Detection	
3.5 Machine Learning Detection	
3.6 Decision and Alert System	
CHAPTER 4: RESULTS AND DISCUSSION	
4.1 Experimental Setup	
4.2 Results	
4.3 Discussion	
CHAPTER 5: CONCLUSION	
5.1 Conclusion	
5.2 Future Work	
References	
Team Details	

Chapter 1: Introduction

1.1 Introduction

As the internet grows, so do cyber threats. Organizations across the world need reliable systems to protect their networks from unauthorized access and malicious activity. An Intrusion Detection System (IDS) is a tool that monitors network traffic and raises alerts when it detects something suspicious.

There are two main types of IDS. Signature-based systems compare traffic against a list of known attack patterns — they work well for familiar threats but cannot detect new attacks. Anomaly-based systems learn what normal traffic looks like and flag anything unusual — they can catch unknown threats but often raise too many false alarms. A Hybrid IDS combines both methods to get better results than either approach alone.

1.2 Problem Statement

Existing IDS solutions have two major problems: signature-based systems miss new (zero-day) attacks, while anomaly-based systems generate too many false alerts. This wastes analyst time and allows real threats to go unnoticed. Additionally, most existing hybrid systems are too complex, too slow, or have never been tested on live networks.

This project addresses these problems by building a simple, efficient hybrid IDS that combines both detection methods and uses a confidence filter to reduce false alarms.

1.3 Objectives

The main objectives of this project are:

- To design a hybrid IDS that combines signature matching with machine learning
- To train and test a Random Forest and XGBoost ensemble on the UNSW-NB15 dataset
- To reduce the number of false positive alerts using a confidence threshold
- To select the most important features for faster and lighter detection
- To evaluate and compare the system against existing approaches

1.4 Scope of Research

This project covers network-level intrusion detection using flow-based traffic features. It does not include deep packet inspection, analysis of encrypted traffic, or live deployment on enterprise networks. Those areas are left for future work.

Chapter 2: Literature Review

2.1 Background

Intrusion detection research started in the late 1980s with simple rule-based expert systems. Over the years, datasets like KDD Cup, NSL-KDD, UNSW-NB15, and CICIDS were created to benchmark different IDS methods. Research has moved from basic classifiers (Decision Trees, Naive Bayes) to more advanced deep learning (CNN, LSTM) and ensemble methods (Random Forest, XGBoost).

2.2 Review of Related Work

Several researchers have proposed hybrid IDS approaches in recent years. The table below summarizes key papers reviewed for this project:

Table 2.1: Summary of Related Work

No.	Author	Method	Dataset	Accuracy	Limitation
1	Henry et al., 2023	CNN + GRU	CICIDS-2017	98.73%	Poor minority class detection
2	Sun et al., 2020	CNN + LSTM	CICIDS-2017	98.67%	Rare attack issues
3	Sajid et al., 2024	XGBoost + CNN + LSTM	Multiple	98.55%	High computational cost
4	Liu et al., 2021	K-Means + RF + CNN	NSL-KDD, CICIDS	99.91%	Too slow for real-time use
5	Verma et al., 2025	PSO + Ensemble ML	NSL-KDD, CICIDS2018	99.83%	No real-time validation
6	Maseno et al., 2022	Systematic Review	Multiple	N/A	No model proposed

Most of the reviewed systems achieve high accuracy (98–99%) but come with major limitations — they are tested only on offline datasets, do not report false positive rates, and are too computationally heavy for practical real-time use.

2.3 Research Gaps

Based on the review, the following key gaps were identified:

- Most systems do not measure or reduce false positive rates
- Deep learning models are too slow for real-time deployment on normal hardware
- Almost no system has been tested on live network traffic — only offline datasets
- There is no standard framework for combining signature and anomaly detection

This project directly addresses these gaps through a lightweight ML model, confidence-based filtering, and a modular hybrid pipeline.

Chapter 3: Methodology

3.1 System Overview

The proposed system works as a pipeline. Network traffic is first captured and converted into feature data. This data is then checked against known attack signatures. If no match is found, the machine learning model analyzes it. The outputs from both checks are combined in a Decision Engine, which produces final alerts for the user.

3.2 Data Used

The system was trained and tested using the UNSW-NB15 dataset, a widely used benchmark for network intrusion detection. It contains over 2.5 million network flow records with 9 different types of attacks. 80% of the data was used for training and 20% for testing.

3.3 Data Preprocessing

Before training, the raw data was cleaned and prepared through the following steps:

- Missing values were filled using average values for numbers and most common values for text fields
- Text-based features (like protocol type) were converted into numbers the model can understand
- All values were scaled to a standard range to prevent large numbers from overpowering the model
- Rare attack types were oversampled to make training more balanced (using SMOTE)
- Only 9 key features were selected from the original 49 — called the Power 9 features

Table 3.1: Power 9 Features Used for Detection

Feature	What It Measures	Type
dur	How long the network flow lasted	Flow Duration
spkts	Number of packets sent from source to destination	Packet Volume
dpkts	Number of packets sent from destination to source	Packet Volume
sbytes	Total data (bytes) sent from source to destination	Byte Volume
dbytes	Total data (bytes) sent from destination to source	Byte Volume
sttl	Time-To-Live value from the source IP	TTL Dynamics
dttl	Time-To-Live value from the destination IP	TTL Dynamics
sload	Data sent per second from source (bits/sec)	Traffic Load
dload	Data sent per second from destination (bits/sec)	Traffic Load

These 9 features capture the most important differences between normal traffic and attacks, such as how much data is being sent and how fast.

3.4 Signature-Based Detection

This part of the system checks incoming traffic against a database of known attack patterns. If a match is found, the system immediately raises a Known Attack Alert. Traffic that does not match any known pattern is passed to the machine learning module for further analysis. This step is fast and uses very little computing power.

3.5 Machine Learning Detection

The machine learning module uses two models working together:

- Random Forest — builds many decision trees and combines their votes. It is good at handling imbalanced data.
- XGBoost — builds trees one at a time, with each tree learning from the mistakes of the previous one. It performs well on structured data like network flow records.

Both models are combined using weighted voting — XGBoost gets 60% weight and Random Forest gets 40%. Their combined score is used to decide if traffic is an attack or not. A confidence threshold of 0.90 is applied, meaning traffic is only flagged as an attack if both models are highly confident. This greatly reduces false alarms.

3.6 Decision and Alert System

The Decision Engine collects alerts from both the signature module and the ML module. It removes duplicate alerts (e.g., the same source IP flagged multiple times in 60 seconds), ranks alerts by severity, and sends the final output to a web dashboard, a real-time alert console, and an audit log. Security teams see only the most important, non-duplicate alerts.

Chapter 4: Results and Discussion

4.1 Experimental Setup

All experiments were carried out using Python 3.10 with the scikit-learn and XGBoost libraries. The UNSW-NB15 dataset was used for training and testing. The data was split 80/20 (train/test) and all preprocessing was done only on the training data to keep the test results unbiased.

4.2 Results

Table 4.1: Final Performance Results

Model	Accuracy	Precision	Recall	F1-Score	False +ve Rate
Random Forest (standalone)	89.7%	88.4%	87.2%	87.8%	6.3%
XGBoost (standalone)	91.2%	90.1%	89.5%	89.8%	4.8%
Proposed Hybrid System	94.6%	95.8%	92.3%	94.0%	1.2%

4.3 Discussion

The proposed hybrid system clearly outperforms both individual models. The key highlights are:

- Accuracy improved from 91.2% (XGBoost alone) to 94.6% with the hybrid approach
- The false positive rate dropped dramatically from 4.8% to just 1.2% — meaning far fewer false alarms
- Precision of 95.8% means nearly 96 out of every 100 alerts raised are genuine attacks
- Using only 9 features instead of 49 made the system 4.7 times faster without losing much accuracy

Compared to deep learning-based approaches, our system is simpler and runs on ordinary hardware, while still delivering competitive detection performance. Other methods in the literature achieve higher accuracy numbers, but they do not report their false positive rates — which is the most critical metric for real-world use.

Chapter 5: Conclusion

5.1 Conclusion

This project successfully built a Hybrid Intrusion Detection System that combines signature-based and machine learning-based detection. The system was tested on the UNSW-NB15 dataset and achieved 94.6% accuracy with a false positive rate of only 1.2%.

The key outcomes of this project are: the hybrid approach works better than using either method alone; applying a confidence threshold significantly reduces false alarms; and using just 9 carefully selected features keeps the system fast and lightweight without sacrificing accuracy.

The system is practical, easy to deploy, and does not need expensive hardware — making it suitable for real-world use in small and medium-sized organizations.

5.2 Future Work

There are several ways this system can be improved in the future:

- Test on live network traffic instead of just offline datasets
- Automatically adjust the confidence threshold based on real-time conditions
- Add support for analyzing encrypted network traffic
- Connect the system with enterprise security platforms like Splunk or IBM QRadar
- Test on additional datasets (CICIDS2017, NSL-KDD) for broader validation

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